Data Exercise

Cleaning, Structure and Formatting & Data Entry Format

Goal

Get raw data and give the steps of cleaning, displaying my ability to do so.

Data

Kaggle Dirty Cafe Sales

GitHub Repository

Link to GitHub Repository

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Steps for data cleaning

1. Duplicate raw_data (for safety).
2. Checked in clean_data sheet, trimmed spaces.
3. In clean_data sheet, replaced columns with ERROR (1,622) or UNKNOWN (1,634) cells with blank cells.
4. Price per unit could be filled, as Items have a fixed price, put prices in a new sheet and then for blank Price per unit cells, I did a VLOOKUP formula.
   a. =VLOOKUP(B58,item_price!$A$1:$B$9,2,false)
   b. Cell (cell to be matched, range of item price table, index of column if there is a match, false=exact match).
5. Total Spent column, fill it with calculation C*D cells, Quantity and Price per unit could be also filled in case there are other 2 cells available with data, otherwise, left them blank.

Steps for data cleaning (with arrayformula)

1. Duplicate raw_data (for safety).
2. Copy and paste headers, copy ID values (check for unique values)
3. Create formula
   a. Item column:
   ```
   =ARRAYFORMULA(IF(raw_data!A2:A="","",
   IF(REGEXMATCH(UPPER(raw_data!B2:B&""),"^(ERROR|UNKNOWN)?$"
   ),"",PROPER(TRIM(raw_data!B2:B&""))))
   ))
   ```
   b. Quantity column:
   ```
   =ARRAYFORMULA(IF(raw_data!A2:A="","",
   IF(REGEXMATCH(UPPER(raw_data!C2:C&""),"^(ERROR|UNKNOWN)?$")
   ,"",IFERROR(VALUE(raw_data!C2:C),""))
   ))
   ```

Steps for data cleaning (with arrayformula)

3. Create formula
    a. Price per unit column:

```
=ARRAYFORMULA(IF(raw_data!A2:A="","",
 IF(
   REGEXMATCH(UPPER(raw_data!D2:D&""),"^(ERROR|UNKNOWN)?$"),
   IFERROR(VLOOKUP(clean_data!B2:B,price_map!A:B,2,FALSE),""),
   IFERROR(VALUE(raw_data!D2:D),"")
 )
))
```

    b. Total spent column:

```
=ARRAYFORMULA(IF(raw_data!A2:A="","",
 IF(
   REGEXMATCH(UPPER(raw_data!E2:E&""),"^(ERROR|UNKNOWN)?$"),
   IF(clean_data!C2:C*clean_data!D2:D=0,"",clean_data!C2:C*clean_data!D2:D),
   IFERROR(VALUE(raw_data!E2:E),"")
 )
))
```

Steps for data cleaning (with arrayformula)

3. Create formula
    a. Payment method column:
    =ARRAYFORMULA(IF(raw_data!A2:A="","",
    IF(REGEXMATCH(UPPER(raw_data!F2:F&""),"^(ERROR|UNKNOWN)?$"),"",PROPER(TRIM(raw_data!F2:F&""))))
    ))

    b. Location column:
    =ARRAYFORMULA(IF(raw_data!A2:A="","",
    IF(REGEXMATCH(UPPER(raw_data!G2:G&""),"^(ERROR|UNKNOWN)?$"),"",PROPER(TRIM(raw_data!G2:G&""))))
    ))
    c. Transaction column:
    =ARRAYFORMULA(IF(raw_data!A2:A="","",
    IF(REGEXMATCH(UPPER(raw_data!H2:H&""),"^(ERROR|UNKNOWN)?$"),"",IFERROR(DATEVALUE(raw_data!H2:H),""))
    ))

Steps for data cleaning (with arrayformula)

4. After cleaning data, copied and paste values only and later filled the missing values with VLOOKUP formula.

After cleaning, we can format the database to make it more structured.
If it's being used as data entry, we can create conditional formatting to be able to make the data easier to write.

Steps for data entry

I'm able to create a google sheets or excel file that you can edit for data entry.

For columns color red: You have to entry data or just move the cell.
For columns color blue: From data entered, it calculates automatically.

Some extra clarifications:
For ID, a random number is generated, you can copy and paste the value to keep it or if you prefer, you can just write manually the id number.
For date, you can just drag the cell down to generate the date from today, it will display it automatically.
For other columns, there was some data validation rules created so you could only select an specific value (this reduces human error like [sandwich|Sandwich|sanduick] because you will be only be able to put the correct value.

	A	B	C	D	E	F	G	H
1	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
2	TXN_9812367	Salad	4	5	20	Credit Card	In-Store	04/03/2026
3	TXN_6754974	Sandwich	6	4	24	Digital Wallet	Takeaway	04/03/2026
4	TXN_9269596	Coffee	1	2	2	Cash	In-Store	04/03/2026
5	TXN_4373920	Cookie	2	1	2	Digital Wallet	In-Store	04/03/2026
6	TXN_5507017	Tea	4	1.5	6	Digital Wallet	In-Store	04/03/2026
7	TXN_7178022	Salad	7	5	35	Credit Card	Takeaway	04/03/2026
8	TXN_5587354	Juice	1	3	3	Cash	Takeaway	04/03/2026
9					0			